IS 547 Course Project – Progress Report

GitHub Link: https://github.com/jazzduvv-56/IS547_Course_Project

Project Summary

For this self-directed project, I set out to understand how two different NOAA systems describe extreme weather in Illinois from 2010 to 2025. Daily instrumental measurements from weather stations (temperature, precipitation, snowfall) and county-level reports of severe events (blizzards, floods, excessive heat) both describe the same physical conditions, but they use different structures, spatial units, temporal granularity, and classification rules. The guiding question is: How well do daily station measurements correspond to officially reported severe weather events, and where do clear inconsistencies appear when we try to align them? Rather than trying to prove that snow causes blizzards, I focused on the curation challenges of harmonizing these two systems.

The final outcome is a clean, reproducible integrated dataset that makes these differences visible. This dataset could later support regional climate risk analysis or policy work. The emphasis is entirely on curation - cleaning, spatial and temporal alignment, documentation, and transparency.

Datasets

I used two official NOAA datasets.

The primary dataset is the Global Historical Climatology Network – Daily (GHCN-Daily). I selected ten Illinois stations that provide good geographic coverage across the North, Central, and Southwest parts of the state: Chicago O'Hare, Chicago Midway, Waukegan, Rockford, Moline, Peoria, Champaign-Urbana, Decatur, Joliet, and Belleville. Key variables include TMAX, TMIN, TAVG, PRCP, SNOW, and SNWD, along with quality and measurement flags.

The second dataset is the NOAA Storm Events Database (Illinois subset, 2010–2025). I limited it to four high-impact event types that align directly with GHCN variables: Winter Storm, Blizzard, Flood, and Excessive Heat. These two sources come from the same trusted agency but structure and represent weather very differently.

Original Curation Goals

In my project plan I outlined three main phases: acquisition and cleaning of GHCN-Daily, acquisition and cleaning of Storm Events, and integration via spatial and temporal harmonization. I planned to follow the DCC Curation Lifecycle Model (Higgins, 2008) and deliver cleaned datasets, a station-to-county mapping table, a final integrated daily climate-and-events dataset, a complete data dictionary, a README, and reproducible Python code. The goal was not statistical modeling but transparent documentation of alignment and divergence between the two reporting systems.

Status of Deliverables

I have now completed all the deliverables outlined in my preliminary plan.

I downloaded and cleaned the full GHCN-Daily records for the ten stations, performed structural, semantic, and context-enrichment cleaning, added REGION and SEASON columns, and created a station-to-county lookup table.

For Storm Events I combined all yearly files, filtered to Illinois only, kept only the four target event types, expanded multi-day events to daily rows, and cleaned damaged fields.

I then performed a left join on DATE and COUNTY to produce the final integrated dataset (final_integrated_illinois_climate_events.csv). All code is fully reproducible and includes detailed comments explaining every decision.

Justifications of Changes in Scope

I made one small, justified change to the plan. Originally, I planned to use county-boundary shapefiles for a spatial join, as the professor suggested. However, because my ten stations are fixed airport locations, a simple station-to-county lookup table proved more accurate and reproducible. This simplification actually improved data quality while still meeting the spirit of spatial harmonization. I documented the mapping decisions in a reference table and kept the pipeline transparent, which aligns with course concepts of provenance tracking and responsible reuse.

Evidence of Progress Through Artifacts

I have produced substantial, well-documented evidence of curation activities. The artifacts include archived raw NOAA downloads for provenance, structured processing logs, the cleaned GHCN dataset with county and season columns, the cleaned and daily-expanded Storm Events dataset, the station-to-county mapping table, and the final integrated dataset. I also created a complete data dictionary and a detailed README with usage examples. All files are organized in open CSV format with UTF-8 encoding. These artifacts clearly link back to the planned tasks in my project plan and demonstrate meaningful progress in quality assessment, metadata work, and harmonization.

I am on schedule with the timeline I proposed. The next steps in April are to finalize the documentation, perform a reproducibility test from the raw files, run basic mismatch statistics, and prepare the final report. Overall, this project has given me hands-on experience with the real-world challenges of harmonizing heterogeneous environmental datasets while maintaining transparency and reusability.

References

Higgins, S. (2008). The DCC Curation Lifecycle Model. *International Journal of Digital Curation, 3*(1), 134–140. https://doi.org/10.2218/ijdc.v3i1.48

Menne, M. J., Durre, I., Vose, R. S., Gleason, B. E., & Houston, T. G. (2012). An overview of the Global Historical Climatology Network-Daily Database. *Journal of Atmospheric and Oceanic Technology, 29*(7), 897–910. https://doi.org/10.1175/JTECH-D-11-00103.1

National Centers for Environmental Information. (n.d.). *Global Historical Climatology Network daily (GHCNd)*. NOAA. https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily

National Centers for Environmental Information. (n.d.). *Storm Events Database*. NOAA. https://www.ncei.noaa.gov/stormevents/

I used Grok to help structure the Python code, debug parsing errors, and organize the wording of this progress report. All curation decisions, code logic, station-to-county mapping, event-type choices, and the core ideas in this report are entirely my own.